UK Atomic Energy Authority

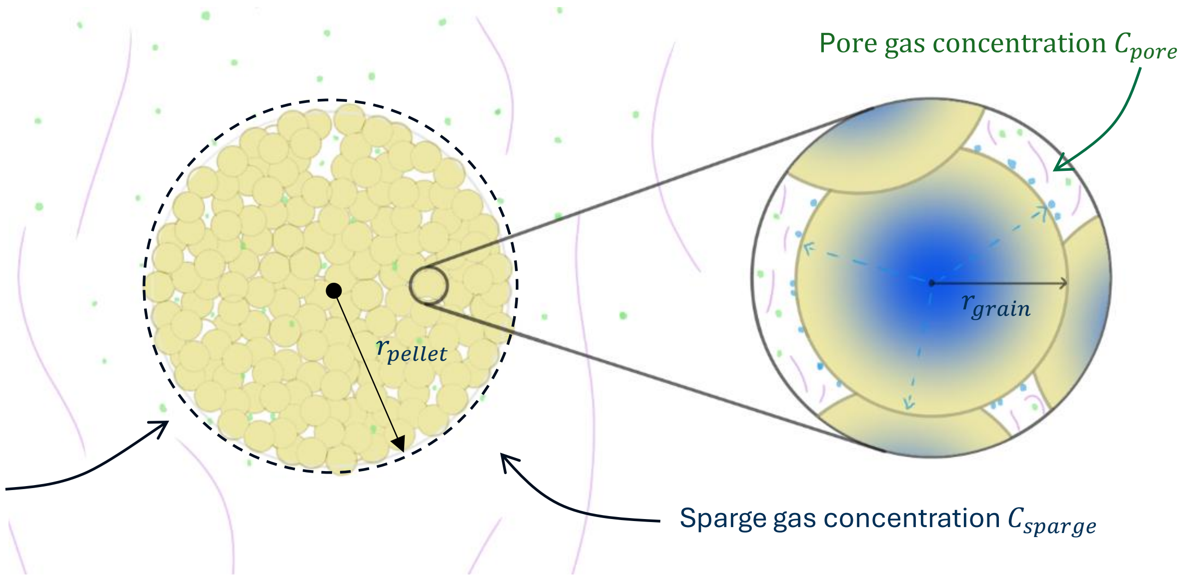
Concept Li2O Pellet Release Model - WIP

Model Formulation

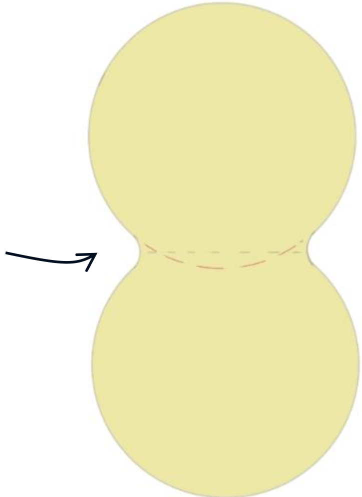
Pellet Structure

Lithium Oxide (Li2O) pellets are typicaly made in a **hot sintering** manufacturing process. This involves taking Li2O powder, pressing into pellets, and then heating to high temperatures to sinter the powder **grains** together and form a solid pellet.

This manufacturing process means the pellets are porous, which is important to consider for a release model.

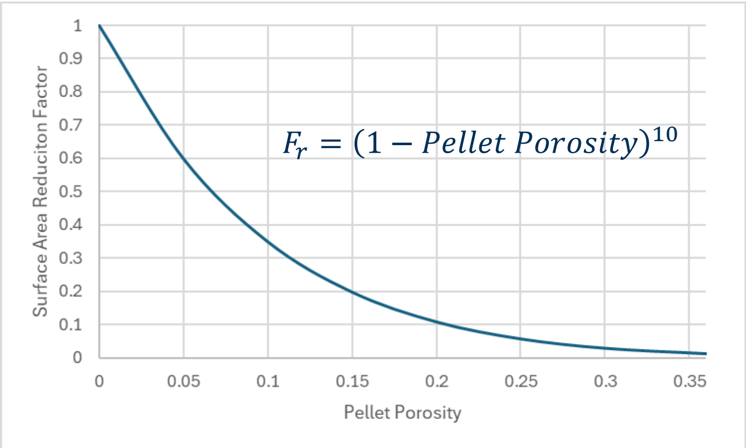


Surface area lost due to sintering necking between grains



For this initial concept model, the pellet is modelled as being made up of equally-sized spherical grains of solid Li2O. It is assumed that the density of the pellet is such that porosity between the grains is open (i.e. there is a direct pathway from the grain interstitials to the pellet boundary/sparge gas). Tritium generated in the grain bulk material diffuses to the surface, desorbs into the slow-moving gas in the pores, and then escapes into the sparge gas which is continually replenished.

The sintering process results in a reduction of exposed surface area of each grain, due to some of the surface 'necking' and fusing with neighbouring grains. This effect is captured and accounted for in the model via a surface area reduction factor F_r.



The factor Fr can be estimated using known limits at 0 and 0.36 pellet porosity. At 0 porosity, the grains are fully fused and there is no surface remaining exposed, so $F_r = 1$. At 0.36 porosity, the grains are randomly packed spheres with no necking, so $F_r = 0$.

Diffusion – Desorption model for Tritium release from a sphere

When a spherical grain of Lithium Oxide is irradiated, Tritium will be generated uniformly throughout the bulk material (assuming the sphere is small compared to the gradient of the local TBR).

To be released from the sphere and into a sparge gas, Tritium must **diffuse** through the bulk material from the generation site to the surface. Once there, it must then **desorb** from the surface into the pore gas. For simplicity, in the solid phase tritium can exist in two states: mobile tritium ($\boldsymbol{C_m}$), and trapped tritium ($\boldsymbol{C_t}$).

The rate of change of mobile tritium at any point in the bulk material is the result of four processes:

$$\frac{\partial C_m}{\partial t} = D\nabla^2 C_m + G - k_t C_m(N_t - C_t) + k_d C_t$$

- $\boldsymbol{D\nabla^2 C_m}$ (Diffusion): transport of mobile tritium through the solid, governed by Fick's second law.
- $\boldsymbol{G}$ (Generation): source term, representing the rate tritium is generated by neutron reactions. This happens randomly throughout the bulk material, so the term is independent of position or concentration.

  The generation rate is the source fluence (τ) multiplied by the volumetric TBR (TBR_{vol}) found via neutronics modelling.

$$G = TBR_{vol} \times \tau$$

- $\boldsymbol{k_t C_m(N_t - C_t)}$ (Trapping): represents the rate at which mobile tritium is trapped at sites in the bulk material. The rate is proportional to the concentration of mobile tritium and the concentration of unoccupied trapping sites.
- $\boldsymbol{k_d C_t}$ (Detrapping): represents the rate at which trapped tritium is released back into the mobile population. Detrapping occurs when a trapped atom is randomly thermally excited such that it exceeds the trapping energy, therefore the frequency of this is proportional to the concentration of atoms that are currently trapped.

The rate of change of trapped tritium concentration reflects the trapping parts of the mobile tritium concentration PDE:

$$\frac{\partial C_t}{\partial t} = k_t C_m (N_t - C_t) - k_d C_t$$

Working in spherical coordinates, the main equation is written as:

$$\frac{\partial C_m}{\partial t} = D\left(\frac{\partial^2 C_m}{\partial r^2} + \frac{2}{r}\frac{\partial C_m}{\partial r}\right) + G - k_t C_m (N_t - C_t) + k_d C_t$$

Once tritium has reached the surface of the grain, it can **desorb** into the surrounding pore gas. For this model it is assumed the desorption rate is a second-order function of the mobile concentration at the surface, due to the recombination requirement for Hydrogen isotope release.

Tritium may also **adsorb** from the surrounding gas onto the surface of the grain. It is assumed this is a first-order function of the concentration tritium in the gas. So, the rate of release from the grain surface (the flux J_{grain}) is:

$$J_{grain} = k_{des} \times {C_{m,\,surface}}^2 - k_{ads} \times C_{pore}$$

The release rate must be equal to the rate of diffusion normal to the surface at the grain boundary.

At the centre of the spherical grain, symmetry means there is no net flux of tritium as the concentration gradient is zero (flat).

These provide two boundary conditions for the main equation:

$$\left.\frac{\partial C_m}{\partial r}\right|_{r=r_{grain}} = -\frac{J_{grain}}{D}$$

$$\left.\frac{\partial C_m}{\partial r}\right|_{r=0} = 0$$

By coupling the rate of diffusion at the surface to the pore gas concentration C_{pore}, it is also linked to the sparge gas concentration C_{sparge}.

Making the approximation that all the gas inside the pore network is at the same concentration, the change in concentration can be modelled as a mass balance equation:

$$\frac{dC_{pore}}{dt} = \underbrace{J_{grain} \times A_{internal}}_{release\ from\ grains} - \underbrace{J_{pellet} \times A_{external}}_{transfer\ to\ sparge}$$

Where $A_{internal}$ is the internal exposed surface area of the grains / pore network, and $A_{external}$ is the external boundary surface area of the pellet.

$A_{internal}$ is calculated as:

Solid volume of pellet

$$A_{internal} = F_r \times \frac{\left(\frac{4}{3} \times \pi \times {r_{pellet}}^3\right) \times \left(1 - Porosity_{pellet}\right)}{r_{grain}}$$

Multiply be a surface reduction factor to account for grain surface area lost due to sintering necking.

Divide by the characteristic length of a grain to get an estimate of surface area

$A_{external}$ is the pellet surface area:

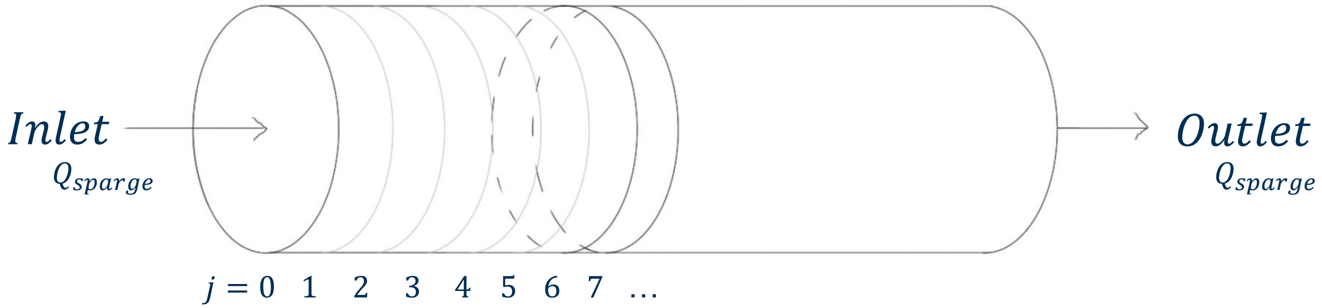
$$A_{external} = 4 \times \pi \times {r_{pellet}}^2$$

The pellet release rate J_{pellet} is modelled as a simple first-order mass transfer rate:

$$J_{pellet} = h_{pellet} \times \left(C_{pore} - C_{sparge}\right)$$

The sparge gas concentration varies both temporally and spatially; the concentration at a point in the packed bed depends both on the duration of release, and on the concentration of the gas upstream of the point of interest. This complex relationship means it is not possible to define the sparge gas concentration as a simple formula of time and position, and it must instead be solved numerically.

The packed bed of pellets has a sparge gas inlet at one end, and an outlet at the other. This enables it to be modelled as a **plug flow reactor**, to find a numeric solution to the sparge gas concentrations at different positions and times.



For plug 0 (at the inlet), the sparge concentration can be modelled as:

$$\frac{dC_{sparge,j=0}}{dt} = \frac{\overbrace{J_{Pellet,j=0} \times A_{pellets,plug}}^{Release\ in\ from\ pellets} - \overbrace{Q_{sparge} \times C_{sparge,j=0}}^{Removal\ via\ sparge\ flow}}{V_{sparge,\,plug}}$$

Assuming the sparge gas entering at the inlet has a concentration $C_{sparge} = 0$.

For downstream plugs, the sparge concentration must consider the upstream release and can be modelled as:

$$\frac{dC_{sparge,j=n}}{dt} = \frac{\overbrace{J_{Pellet,j=n} \times A_{pellets,plug}}^{Release\ in\ from\ pellets} + \overbrace{Q_{sparge} \times C_{sparge,j=n-1}}^{Upstream\ concentration\ in} - \overbrace{Q_{sparge} \times C_{sparge,j=n}}^{Removal\ via\ sparge\ flow}}{V_{sparge,\,plug}}$$

The concentrations in the solid phase and in the pore gas will also vary with position j through the packed bed.

Full set of model equations

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Mobile tritium concentration in bulk material:

$$\frac{\partial C_m}{\partial t} = D\left(\frac{\partial^2 C_m}{\partial r^2} + \frac{2}{r}\frac{\partial C_m}{\partial r}\right) + G - k_t C_m (N_t - C_t) + k_d C_t$$

$$\left.\frac{\partial C_m}{\partial r}\right|_{r=r_{grain}} = -\frac{J_{grain}}{D}$$

$$\left.\frac{\partial C_m}{\partial r}\right|_{r=0} = 0$$

Trapped concentration in bulk material:

$$\frac{\partial C_t}{\partial t} = k_t C_m (N_t - C_t) - k_d C_t$$

Pore gas concentration:

$$\frac{dC_{pore,j=n}}{dt} = \underbrace{J_{grain,j=n} \times A_{internal}}_{release\ from\ grains} - \underbrace{J_{pellet,j=n} \times A_{external}}_{transfer\ to\ sparge}$$

$$J_{grain} = k_{des} \times C_{m,\,surface}{}^2 - k_{ads} \times C_{pore}$$

$$J_{pellet} = h_{pellet} \times \left(C_{pore} - C_{sparge}\right)$$

$$A_{internal} = F_r \times \frac{\left(\frac{4}{3} \times \pi \times r_{pellet}{}^3\right) \times \left(1 - Porosity_{pellet}\right)}{r_{grain}}$$

$$A_{external} = 4 \times \pi \times r_{pellet}{}^2$$

For plug 0 (at the inlet), the sparge concentration can be modelled as:

$$\frac{dC_{sparge,j=0}}{dt} = \frac{\overbrace{J_{pellet,j=0} \times A_{pellets,plug}}^{release\ in\ from\ pellets} - \overbrace{Q_{sparge} \times C_{sparge,j=0}}^{removal\ via\ sparge\ flow}}{V_{sparge,\,plug}}$$

$A_{pellets,plug} = Total\ external\ surface\ area\ of\ pellets\ per\ plug$

$Q_{sparge} = Volumetric\ sparge\ flow\ rate$

$V_{sparge,plug} = Volume\ of\ sparge\ gas\ per\ plug$

Assuming the sparge gas entering at the inlet has a concentration $C_{sparge} = 0$.

For downstream plugs, the sparge concentration must consider the upstream release and can be modelled as:

$$\frac{dC_{sparge,j=n}}{dt} = \frac{\overbrace{J_{pellet,j=n} \times A_{pellets,plug}}^{release\ in\ from\ pellets} + \overbrace{Q_{sparge} \times C_{sparge,j=n-1}}^{upstream\ concentration\ in} - \overbrace{Q_{sparge} \times C_{sparge,j=n}}^{removal\ via\ sparge\ flow}}{V_{sparge,\,plug}}$$

Numeric Implementation

Testing in Python

To test the model in python, we can approximate the continuous PDEs with a discretised numerical method. The PDE gives the *instantaneous* rate of change over time, this can be approximated by the change over a relatively small time step Δt.

$$\frac{\partial C_m}{\partial t} = \frac{\text{Future Concentration} - \text{Current Concentration}}{\text{Time Step}} = \frac{C_m(t+\Delta t) - C_m(t)}{\Delta t}$$

Substituting this equation back into the PDE:

$$\frac{C_m(t+\Delta t) - C_m(t)}{\Delta t} = D\nabla^2 C_m + G - k_t C_m(N_t - C_t) + k_d C_t$$

Rearranging to isolate the future concentration term:

$$C_m(t+\Delta t) = C_m(t) + \Delta t(D\nabla^2 C_m + G - k_t C_m(N_t - C_t) + k_d C_t)$$

The continuous terms on the right now need to be replaced by discrete equivalents. The spherical pellet is discretised as a set of N radial nodes with spacing Δr.

When using spherical coordinates, the mobile concentration curvature is:

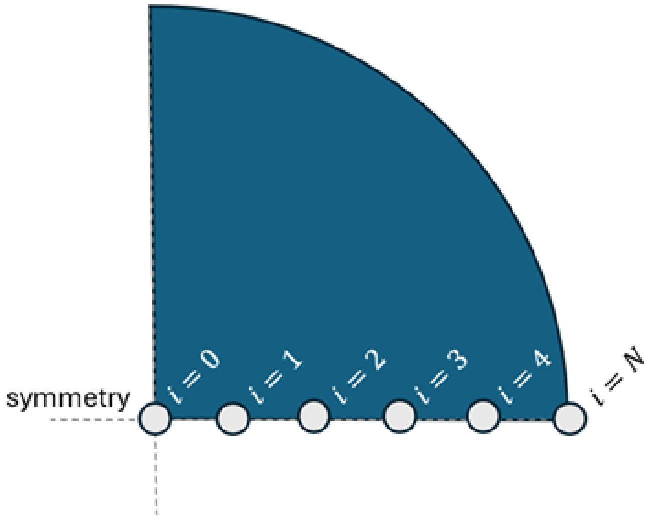
$$\nabla^2 C_{m,i} = \frac{\partial^2 C_m}{\partial r^2}(\text{standard term}) + \frac{2}{r_i} \times \frac{\partial C_m}{\partial r}(\text{spherical correction term})$$

In numeric form, this is:

$$\nabla^2 C_{m,i} = \frac{C_{m,i+1} - 2C_{m,i} + C_{m,i-1}}{\Delta r^2} + \frac{1}{r_i}\left(\frac{C_{m,i+1} - C_{m,i-1}}{\Delta r}\right)$$

At the centre of the sphere (node 0), the equation is modified to:

$$\nabla^2 C_{m,i=0} = 6 \times \frac{(C_{m,i=1} - C_{m,i=0})}{\Delta r^2}$$



The rate of diffusion is defined as:

$$\left.\frac{\partial C_m}{\partial t}\right|_{\text{diffusion}} = D\nabla^2 C_m$$

In spherical coordinates this is:

$$\left.\frac{\partial C_m}{\partial t}\right|_{\text{diffusion}} = D\nabla^2 C_m = D\left(\frac{\partial^2 C_m}{\partial r^2} + \frac{2}{r_g}\frac{\partial C_m}{\partial r}\right)$$

At the grain surface, we enforce the boundary condition:

$$J_{grain} = -D\frac{\partial C_m}{\partial r} \text{ at } r = r_g, \rightarrow \frac{\partial C_m}{\partial r} = -\frac{J_{grain}}{D}$$

Discretising the formula for the second derivative yields:

$$\left.\frac{\partial^2 C_m}{\partial r^2}\right|_{i=N} \approx \frac{C_{N+1} - 2C_N + C_{N-1}}{(\Delta r)^2}$$

The concentration at C_{N+1} is a ‘ghost point’ beyond the surface of the grain. We can define this ghost point by applying the boundary condition. This first derivative at the boundary can be approximated as:

$$\left.\frac{\partial C_m}{\partial r}\right|_{i=N} \approx \frac{C_{N+1} - C_{N-1}}{2\Delta r}$$

The defines the diffusion flux over the boundary. Equating this with the known boundary condition:

$$\frac{C_{N+1} - C_{N-1}}{2\Delta r} = -\frac{J_{grain}}{D}$$

Solving for the ghost point C_{N+1}:

$$C_{N+1} = C_{N-1} - \frac{2\Delta r \times J_{grain}}{D}$$

Substituting this back into the second derivative formula:

$$\left.\frac{\partial^2 C_m}{\partial r^2}\right|_{i=N} \approx \frac{\left(C_{N-1} - \frac{2\Delta r \times J_{grain}}{D}\right) - 2C_N + C_{N-1}}{(\Delta r)^2} = \frac{2C_{N-1} - 2C_N}{(\Delta r)^2} - \frac{2\Delta r \times J_{grain}}{D(\Delta r)^2} = \frac{2(C_{N-1} - C_N)}{(\Delta r)^2} - \frac{2 \times J_{grain}}{D\Delta r}$$

We have expressions for both of the derivatives required to define the diffusion at the boundary, with the J_{grain} boundary condition enforced. Plugging these back into the original diffusion equation:

$$\frac{\partial C_m}{\partial t}_{\text{diffusion}} = D\left(\frac{\partial^2 C_m}{\partial r^2} + \frac{2}{r_g}\frac{\partial C_m}{\partial r}\right) = D\left(\left[\frac{2(C_{N-1} - C_N)}{(\Delta r)^2} - \frac{2 \times J_{grain}}{D\Delta r}\right] + \frac{2}{r_g}\left[-\frac{J_{grain}}{D}\right]\right)$$

$$\frac{\partial C_m}{\partial t}_{\text{diffusion}} = \left(D\frac{2(C_{N-1} - C_N)}{(\Delta r)^2} - D\frac{2 \times J_{grain}}{D\Delta r} - D\frac{2 \times J_{grain}}{Dr_g}\right)$$

$$\frac{\partial C_m}{\partial t}_{\text{diffusion}} = D\frac{2(C_{N-1} - C_N)}{(\Delta r)^2} - \frac{2 \times J_{grain}}{\Delta r} - \frac{2 \times J_{grain}}{r_g}$$

$$\frac{\partial C_m}{\partial t}_{\text{diffusion, surface}} = \underbrace{D\frac{2(C_{N-1} - C_N)}{(\Delta r)^2}}_{\text{Diffusion from interior}} \underbrace{-2 \times J_{grain}\left(\frac{1}{\Delta r} + \frac{1}{r_g}\right)}_{\text{Release from surface}}$$

This tells us the rate of diffusion and release at the grain surface. However, the grain model assumes a perfect spherical grain. In reality, some surface area is lost due to necking between grains that happens during sintering. To account for this, we scale the release term by the surface area reduction factor F_r.

$$\frac{\partial C_m}{\partial t}_{\text{diffusion, surface}} = \underbrace{D\frac{2(C_{N-1} - C_N)}{(\Delta r)^2}}_{\text{Diffusion from interior}} \underbrace{-F_r \times 2 \times J_{grain}\left(\frac{1}{\Delta r} + \frac{1}{r_g}\right)}_{\text{Effective release from surface}}$$